Southern Cross University

Analgesics, Antipyretics and Anti-inflammatory Agents

IP5A1: Understanding Inflammation and pain

Dr Ken Wojcikowski

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Analgesics, Antipyretics and Anti-inflammatory Agents

IP5A1: Understanding Inflammation and pain



Ken Wojcikowski, PhD (Pharmacology and Toxicology; School of Medicine, University of Queensland)

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Specific objectives for Part 1

- Define the term 'pain' and briefly discuss the most important points about pain from a clinical aspect.
- Briefly describe inflammation, placing the most common anti-inflammatory agents into the context of that description.
- Describe the role of prostaglandins and cytokines in the process of inflammation and fever, and cite examples of their actions. Relate this to the anti-inflammatory agents that affect these mediators of inflammation.
- Briefly describe the pain pathway, overviewing the way in which our analgesics work.
- Describe the gate control theory of pain, and the way in which our understanding of pain has evolved.

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Definition of pain; types of pain; importance of treating pain

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What is pain?

International Association for the study of pain:

"Pain is an unpleasant sensory and emotional experience, associated with actual or potential tissue damage, or described in terms of such damage."

Chironex sp. (box jellyfish)



"small box jellyfish (Chironex fleckeri, possibly yamaguchii)" by gautsch. is licensed under CC BY-SA 2.0.

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What is pain?

- Pain is an emotion and a sensation.
  - Protective mechanism
  - Detects, localises and identifies tissue damaging processors.
- Pain varies between individuals.
  - Subjective experience that varies in individuals
    - *Physical*
    - *Psychological (anxiety, memory)*
    - *Environmental factors*
- Pain varies between individuals and situation e.g. during a battle or childbirth.
- Pain is affected by mental state, anxiety, expectation of pain, past experiences and many other factors

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Types of pain

- Acute Pain:
  - Tissue damage
  - Inflammatory cascade
- Sub-acute phase:
  - Transition from acute to chronic pain
  - 4 month window (aggressive treatment indicated now)
  - Neuroplastic changes to CNS
- Chronic pain:
  - CNS becomes sensitized
  - Responds excessively to stimuli
    - Debilitating, expensive & effects everyone surrounding patient
    - Chronic pain begins to have similarities with neuropathic pain

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Acute verses chronic pain

- Chronic pain decreases level of endogenous opioids in the spinal cord
- Changes in dorsal root ganglion include inflammation that increases transmission of impulses.
- Chronic pain often results in loss of appetite, sleep disturbances and depression
- Substance P is a neurotransmitter that lasts MUCH longer than ideal

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Treatment aims for pain

- Decrease pain and suffering/reduce morbidity
- Improve physical and mental function
- Provide hope and emotional wellbeing
- Help patients reach a point of acceptance with their status

…with the smallest dose (least amount of damage)

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Mechanisms of inflammation and anti-inflammatory agents

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Definition of inflammation

(Merriam Webster)

A local response to cellular injury that is marked by:

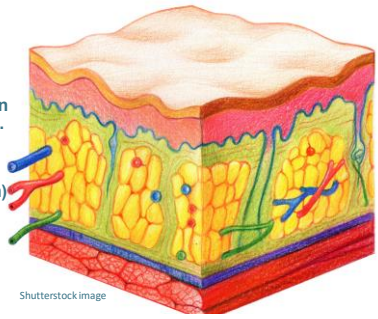
- capillary dilatation,
- leukocytic infiltration,
- redness,
- heat, and
- pain

(Serves as a mechanism initiating the elimination of noxious agents and of damaged tissue).

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Inflammation occurs in the interstitial spaces.

(Between the cells visible in this diagram)

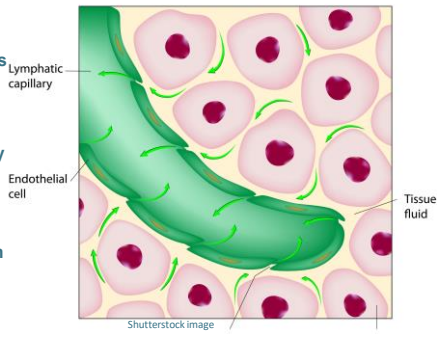


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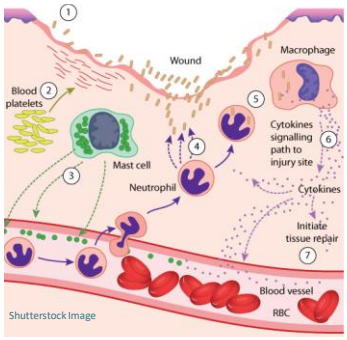
One of the aims of inflammation is to remove the offending microorganism, debris and dead cells and to carry it off into the lymphatic vessels.

This happens in the area between the cells of the organ involved



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1. Bacteria and other pathogens enter wound
2. Platelets from blood release blood-clotting proteins at wound site
3. Mast cells secrete factors that mediate vasodilation and vascular constriction. Delivery of blood, plasma, and cells to injured area increases
4. Neutrophils secrete factors that kill and degrade pathogens
5. Neutrophils and macrophages remove pathogens by phagocytosis
6. Macrophages secrete hormones called cytokines that attract immune system cells to the site and activate cells involved in tissue repair
7. Inflammatory response continues until the foreign material is eliminated and the wound is repaired

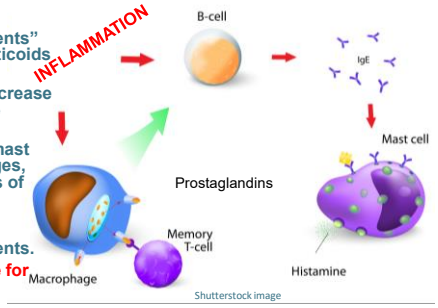
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"Steroidal anti-inflammatory agents" are the glucocorticoids like prednisone.

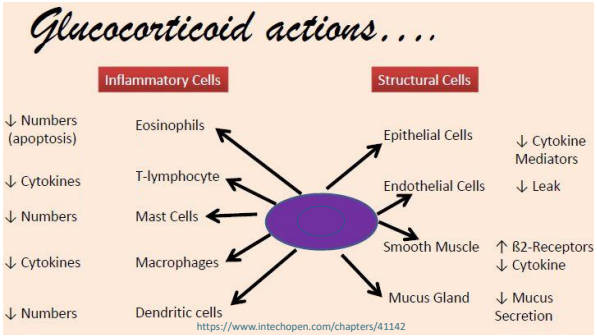
Those agents decrease ALL 'arms' of the immune system.

B cells, T cells, mast cells, macrophages, and all mediators of inflammation are decreased with steroidal anti-inflammatory agents.

See next slide for actions



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Other anti-inflammatory agents

- Other anti-inflammatory drugs target specific aspects of inflammation.

Cyclooxygenase inhibitors

NSAIDS and COX 2 inhibitors target the enzyme that makes prostaglandins.

- NSAIDs (non-steroidal anti-inflammatory drugs that do not act like cortisol and the glucocorticoid steroids).
- COX 2 inhibitors (inhibit cyclooxygenase 2)

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Other anti-inflammatory agents

Some anti-inflammatory agents inhibit the actions of immune cells or block the receptors for inflammatory mediators.

- Mast cell inhibitors, example:
- Leukotriene receptor antagonists.
- Tumour necrosis factor antagonists

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Other anti-inflammatory agents

Agents that have some anti-inflammatory effects but are not considered anti-inflammatory drugs

- Antihistamines (block the histamine receptor)
- Sympathomimetics (stabilize mast cells)

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Mechanisms of pain and analgesics

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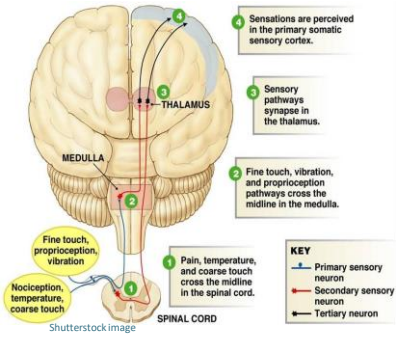
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Overview

The pain pathway consists of three 'orders of neurons'

Depicted here as:

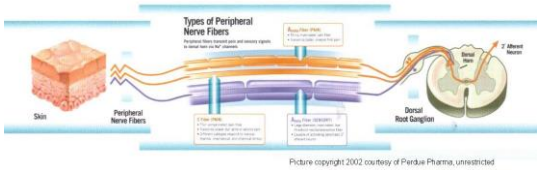
1. Primary
2. Secondary
3. Tertiary



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First order of neurons: the message begins in the periphery, and travels into the spinal cord



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Pain pathways

- A delta fibres
  - Large/myelinated
  - Rapid sharp localised pain
  - Drop off information primarily in the Thalamus
- C fibres
  - Slower/non myelinated
  - Dull, aching poorly localised pain
  - Drop off information primarily in the Brain Stem
  - Chronic pain is primarily through this path

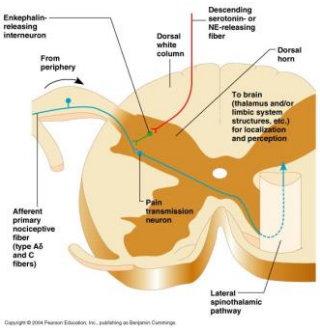
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Modifying the pain

- Stimulation of large, heavily myelinated A beta fibres (which synapse in the same place as A delta fibres) can <u>reduce</u> the pain. These carry mechanical info
  - Thus sucking on the thumb you just crushed or "licking your wounds" decreases pain
- <u>Intensification</u> of pain can occur with higher brain control
  - (fear of the injection)

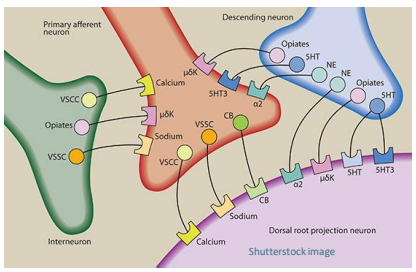
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The 'gate' in the spinal cord



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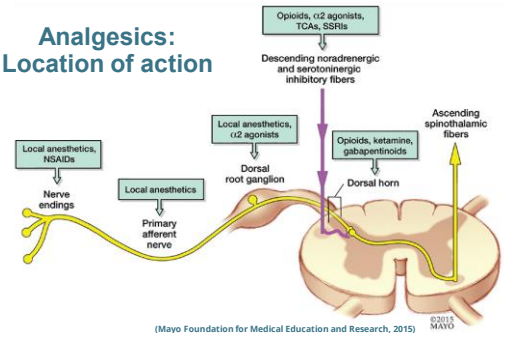
The second 'order of neurons'



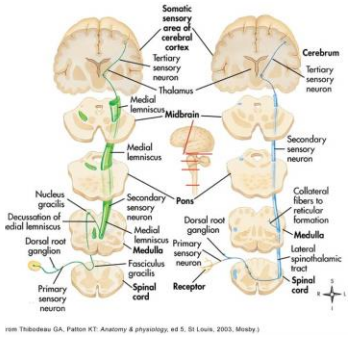
Legend:
5HT/5HT3:Serotonin
CB: cannabinoids
CBZ: carbamazepine
LMT: Lamotrigine
NE: Norepinephrine
SNRI: selective norepinephrine reuptake inhibitor
TCA: tricyclic antidepressant
VSCC: voltage sensitive calcium channel
VSSC: voltage sensitive sodium channel

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Analgesics: Location of action



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rom Thibodeau GA, Patton KT: Anatomy & physiology, ed 5, St Louis, 2003, Mosby.)

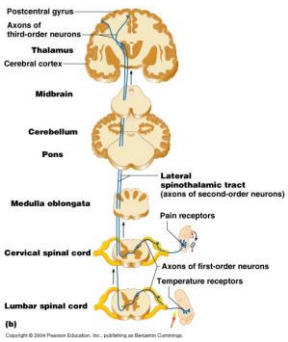
Spinothalamic tract
carries A and C fibres

- The first order of neurons (primary sensory neuron) only goes to the spinal cord before synapsing.
- At the synapse, there are enkephalin (opioid) releasing interneurons from that are controlled by the brain.

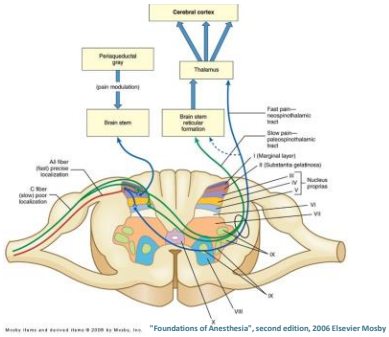
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The second order of neurons travels up the contralateral side, to the thalamus AND BRAINSTEM.

The third order of neurons carries the information to the primary sensory areas of the brain and areas of the limbic system

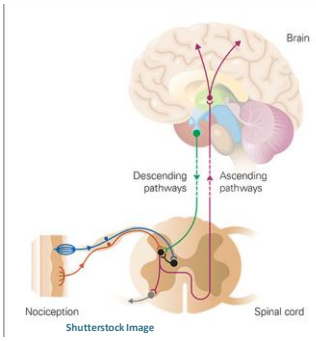


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"Foundations of Anesthesia", second edition, 2006 Elsevier Mosby

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Mechanisms of pain

- Limbic and reticular tracts are involved in alerting body to danger
  - Both regulate the emotional responses to pain
  - Arouse the whole organism to the possibility of damage
  - Autonomic reflexes
  - Neuroendocrine responses

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Analgesics, Antipyretics and Anti-inflammatory Agents
Part 2: NSAIDs, antipyretics, and over the counter analgesics



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Analgesics, Antipyretics and Anti-inflammatory Agents

Part A2: NSAIDs, antipyretics, and over the counter analgesics

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Specific objectives for Part 2

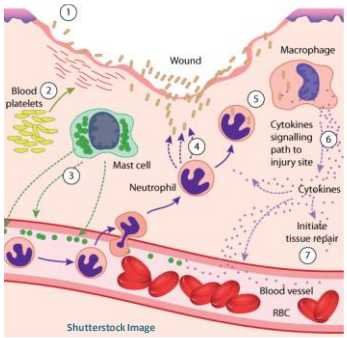
- State the definition of NSAIDS and mechanisms by which NSAIDs work to reduce pain
- Briefly describe fever with respect to the way in which it develops, and how our antipyretic agents work to reduce fever.
- Name at least 7 benefits of fever.
- Using ibuprofen, aspirin and naproxen as examples of NSAIDs, state the indications contraindications and side effects (including long-term side effects) of NSAIDs.
- State the indications contraindications and side effects (including long-term side effects and toxic effects) of paracetamol.

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Fever:
Mechanisms by which it develops
The value of fever as a defence

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1. Bacteria and other pathogens enter wound
2. Platelets from blood release blood-clotting proteins at wound site
3. Mast cells secrete factors that mediate vasodilation and vascular constriction. Delivery of blood, plasma, and cells to injured area increases
4. Neutrophils secrete factors that kill and degrade pathogens
5. Neutrophils and macrophages remove pathogens by phagocytosis
6. Macrophages secrete hormones called cytokines that attract immune system cells to the site and activate cells involved in tissue repair
7. Inflammatory response continues until the foreign material is eliminated and the wound is repaired

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Fever

- A normal adaptive response to endogenous or exogenous pyrogens
- A trigger of the fever, called a pyrogen, crosses the BBB and causes a release of prostaglandin E2 (PGE2) at the hypothalamus. PGE2 then in turn acts on the hypothalamus, which generates a systemic response back to the rest of the body, causing heat-creating effects to match a new temperature level.
- Exogenous pyrogens
  - lipopolysaccharide complex in the cell wall of gram-positive bacteria (mediated by secondary inflammatory molecules)
- Endogenous pyrogens
  - Inflammatory cytokines (Interleukin 1 and 6, tumour necrosis factor and interferon gamma) break into the BBB, signalling the release of prostaglandin E2 which acts on the hypothalamus

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Antipyretics

Before giving antipyretics, please be aware of the benefits of fever:

- Increased temperature kills many micro-organisms
- Increased temperature has adverse effects on the growth and replication of others
- Higher body temperatures decrease serum levels of iron, zinc and copper needed for bacterial replication
- Increased temperature prevents viral replication in infected cells
- Fever is one of the inflammatory factors that promotes liver to produce acute-phase proteins
- Heat binds cations necessary for bacterial reproduction
- Heat increases lymphocytic transformation
- Heat increases motility of polymorphonuclear neutrophils
- Phagocytosis is enhanced during fever
- Production of antiviral interferon may be augmented during fever

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Medications with antipyretic actions

- All NSAIDs (Ibuprofen, Naproxen, others)
- Aspirin
- Paracetamol
- Corticosteroids



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NSAIDs and COX2 inhibitors

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NSAIDs / Aspirin

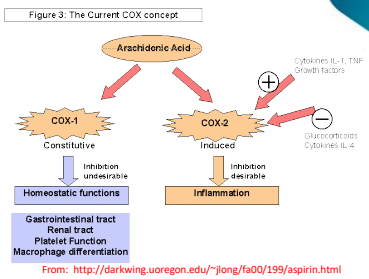
- Mechanism of action
  - Inhibit Prostaglandin formation
    - Inactivation of COX, the enzyme that forms prostaglandins
    - Analgesic
    - Anti-inflammatory
    - Antipyretic
    - E.g. Ibuprofen, Naproxen, indomethacin

Aspirin specific points are discussed in the Cardiovascular portion of this unit



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From: http://darkwing.uoregon.edu/~jlong/fa00/199/aspirin.html

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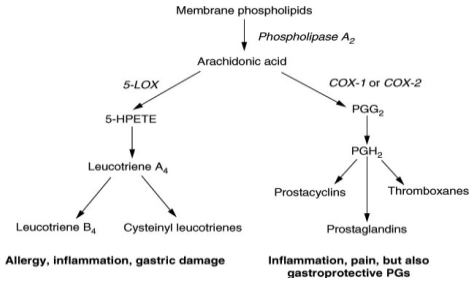
Prostaglandin antagonists

Examples of prostaglandin antagonists are:

- NSAIDs (inhibit cyclooxygenase)
- Corticosteroids (inhibit phospholipase A2 production)
- COX-2 selective inhibitors of cyclooxygenase 2 (…coxib drugs)

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Mediators of inflammation: Prostaglandins and Leukotrienes



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NSAIDs : side effects

- COX1: impaired gastric and renal protection
  - Use a PPI
- CARE:
  - Asthma. (patient-specific: may increase risk of bronchospasm)
  - History of GI bleeding
  - Heart failure, hypertension
    - Na and Fluid retention
  - Renal or hepatic impairment
    - Alcoholic liver disease
  - Cardiovascular risk

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NSAIDs: Warning signs / Risk Factors

- Black stools, Coffee ground vomit (these are from GIT bleeding)
- Age: 65+ (as discussed later)
- History of ulcers
- Long course
- Nephrotoxic drugs or anticoagulants
  - (NSAIDs are contraindicated with Warfarin and may increase nephrotoxicity of other drugs)
- Difficulty breathing

Use lowest dose for shortest time.

- Using a PPI at the same time may decrease the risk of ulcers

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COX 2 Inhibitors (…coxib suffix)

Celecoxib, Parecoxib

- Data from early studies was manipulated

These are no more effective than NSAID's and have serious risks associated with long term use.

Short term use only.

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COX 2 Inhibitors

- Celecoxib, Parecoxib
- No more effective than NSAID's; use if:
  - unresponsive to NSAID's
  - or risk of GI ulceration
- Adverse effects:
  - headache, diarrhoea, abdominal discomfort, acute renal failure & dizziness
- Increased risk of thrombotic events
  - Increased risk of MI/Stroke in studies

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Paracetamol (Acetaminophen)

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Paracetamol

- Said to be the first line treatment for mild-moderate pain (this has been contested in meta analyses determining that it has very little analgesic effect)
- 500mg tablets/ 665mg SR, IV
- Wide usage
- Rapidly absorbed
- Extensive first pass metabolism
- Short half life……1-4 hours (VERY VARIABLE!)

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Paracetamol

- NO known and confirmed mechanism of action:
  - Seems to be converted to the active metabolite in the brain.
  - Little or no effect on peripheral prostaglandin synthesis
  - Unsuitable for reducing inflammation e.g. RA

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Paracetamol toxicity

Paracetamol toxicity is the number one cause of liver failure in Australia, UK, USA, NZ and Canada.

- Toxicity - At risk if:
  - Doses > 4g / day
  - Person metabolizes less quickly (variable half life of 1-4 hours)
  - Client is after the opioid high: Taking combinations of opioids and paracetamol

Paracetamol is often used for suicide attempts: education is necessary! **The more you take, the more TOXIC IT IS. High doses do not increase efficacy!**

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Please reflect on the previous slide

Paracetamol is said to be 'safe' , but it is the #1 cause of liver failure in Australia

28% of cases of liver failure are unintentional

Paracetamol only has 1/140th the analgesic effect of morphine and it cannot get more potent with larger doses

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Paracetamol overdose

- Toxicity if concentration >150mcg/ml/
- Liver toxicity, hypoglycaemia & acute renal tubular necrosis
- May take several days: this is a very painful way to die
- Give N-acetylcysteine
  - Guidelines

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Pain relief in children and the elderly

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Pain relief in children

- Difficult to evaluate
- Under treated & under managed
- Difficult to separate true pain from anxiety







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Analgesics in Children

- Paracetamol
  - 15mg/kg every 6 hours. Max of 90mg/kg/day
  - Safe and well absorbed (30 mins)
- NSAIDS
  - e.g. ibuprofen 8-10mg/kg q8h
- Aspirin…**avoid in children under 18 years due to the possibility of causing Reye's syndrome**
- Codeine
  - Mild - moderate pain
  - 0.5-1mg/kg q4-6 hours. E.g. Painstop

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Cautions regarding children

Dose based on weight (mg/kg)
- If decimal point in wrong place: out by a factor of 10
- Therapeutic duplication
  - e.g. paracetamol containing products

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Factors in the elderly

- Chronic pain increases with age
  - 20% of over 65
  - 50-80% in residential care facilities
- Multiple pathology i.e. several sources of pain
- Behavioural changes may suggest pain
  - e.g. depression, loss of mobility
- Altered sensitivity to drugs
  - e.g. benzodiazepines
- More sensitive to adverse effects
- Multiple therapies and interactions
  - Warfarin with NSAID's and carbamazepine
  - OTC meds

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Factors in the elderly

Pharmacokinetic factors:
- Reduced muscle mass
- Increased fat
- Reduced renal/hepatic function

Above = Increased sensitivity to opioids

Important cautions
- Fluid retention, bleeding and confusion with NSAID's
- Constipation and confusion with opiates
- Start with 25-50% adult dose

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Part 3: Opioids

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Analgesics+ : Part 3 - Opioids



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Specific objectives for Part 3

- Define the terms 'narcotic' and 'opioid'.
- For each of the following opioids, name the most significant contraindications and side effects, the approximate duration of action and the approximate strength relative to each other.
  - *Oxycodone*
  - *Tramadol*
  - *Codeine*
  - *Morphine*
  - Methadone
  - Naloxone (an antagonist)

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Introducing and understanding opioids in general

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Papaver somniferum

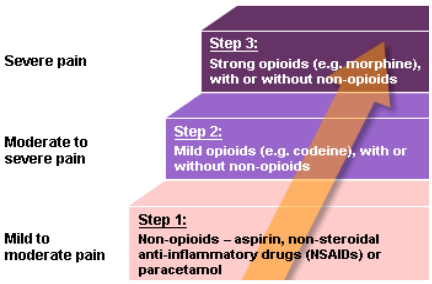




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Analgesic Ladder



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Opioids

- Severe pain: trauma, post op, cancer and severe non cancer related pain
  - Do not abolish pain
- Chronic pain:
  - If other analgesics unsuitable/inadequate
- Cough suppressant
- Diarrhoea
- Prior to anaesthesia

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Opioids

Not all types of pain respond

Risks:
- Tolerance. Mu receptor effect
- Dependence
- Addiction
- Loss of autonomy

- Safe keeping of drugs and medications

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Understanding the actions and side effects of opioids

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Adverse effects of opioids

- Nausea and vomiting
- Constipation (no tolerance)
- Respiratory depression
- Itch
- Pupil constriction
- Drowsiness/sedation
- Urinary retention
- *Dependence/tolerance*
- Hypotension (histamine)

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Routes of administration

- Oral
- Transdermal
- S/c, IV, IM
- CSCI
- PCA
- Nebulised
- Epidural





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Opioid Pharmacology

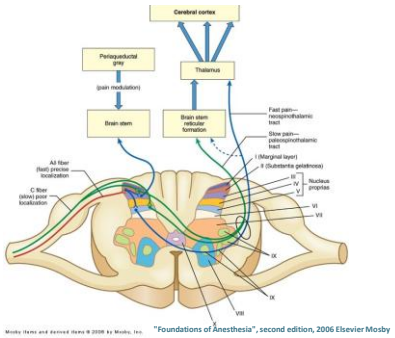
Prevents transmission of pain impulse by acting pre/post synaptically in CNS

Mimics endogenous opioids.

Four opiate receptors in CNS:

- MU: analgesia, respiratory depression
- Kappa: analgesia, sedation
- Delta: hallucinations
- Epsilon: psychosis, dysphoria

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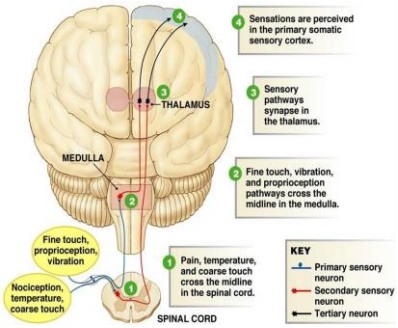
"Foundations of Anesthesia", second edition, 2006 Elsevier Mosby

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The pain pathway consists of three 'orders of neurons'

Depicted here as:

1. Primary
2. Secondary
3. Tertiary



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Opioid Pharmacology

Drug	μ	κ	δ
Agonist			
Morphine	+++	+	+
Methadone	+++		
Pethidine	++	+	+
Fentanil	+++		+
Antagonist			
Naloxone	+++	++	+

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Opiate Pharmacodynamics

EFFECT	mu	kappa	delta
Analgesia	+++	+	+
Euphoria	+++		
Resp depression	+++		++
GIT motility	++	+	++
Physical Dependence	+++	+	
Sedation	++	++	
Pupil Constriction	++	+	

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Specific opioids

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Morphine

- Moderate to severe acute/chronic pain
- Metabolised to glucuronide. M6G. Long half life
- Once 24 hour dose of morphine established, introduce a slow release preparation
- Rescue dose: one 6th of total daily dose
- Schedule 8 drug…..



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Dosing of Morphine

- Titrate dose to response
- Acute pain: 7.5-12.5mg sc/im every 2 hours prn
- Children: weight. 20-40mcg/kg
- 30mg oral morphine equivalent to 10mg sc/im/iv
- Monitor closely



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Opioids for chronic pain

- Many types of chronic pain do not respond to opioids.
- Start: low dose modified release morphine e.g. 5-20mg bd
- Titrate slowly
- Avoid injectable opioids
- Patches:
  - Unsuitable for acute/fluctuating pain
- Codeine :
  - Short acting
  - Mild/moderate pain

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Concerns with Morphine

- Addiction unlikely if used to manage pain…Traditional beliefs
- Tolerance
- Toxicity:
  - Patients with renal impairment
  - Dose increased too rapidly
  - Overlap between therapeutic & toxic concentrations

Monitor regularly for respiratory depression

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Oxycodone

- Oxycodone….greater bioavailability 60-90%
- 1.5-2 times as potent
- Mu and Kappa agonist
- Available as CR and quick release tabs
- Similar tolerability and efficacy
- Hillbilly heroin…chewed oxycontin



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Opioid Induced bowel dysfunction

- Symptoms:
  - Abdominal pain, bloating, reflux, dry, hard stools
- 7 out of 8 patients require laxatives
- Opioid receptors throughout entire gut
  - Decreased peristalsis
  - Decreased secretion of intestinal fluids
  - Increased sphincter tone
  - Enhanced fluid absorption
  - Tolerance does not occur

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Targin: Oxycodone/naloxone

- Reduce opioid induced GI effects:
  - Constipation, nausea.
- 5mg/2.5mg – 40mg/20mg.
- Give twice a day.
- No abuse potential.
- Liver?



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Hydromorphone

- 5-7 times as potent as morphine
- High solubility/Small volume
- Rapid onset and short duration
- Toxic metabolites may accumulate in renal failure….hyperexcitability
- Dose orally every four hours
- Or CSCI

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Codeine

- Weak opioid….Step 2 of WHO ladder
- Metabolised to morphine by CYP2D6
- 10% of Caucasians unable to metabolise
- Can be combined with paracetamol
- Constipation and drowsiness

- *Do you get relief from codeine?*
- That is a question you should ask, because 10% of clients will NOT metabolize the codeine to the active form (morphine)

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Fentanyl

- IV, patch and lozenge
- 100mcg equiv to 10mg morphine s/c
- Patch 25mcg/hr equiv to 90mg/24hrs morphine
  - Avoid in the acute setting
    - Carefully Check patients for patches
- Lozenge for breakthrough
  - Rapid onset 5-15 minutes
    - Lipid soluble
- Lower incidence of constipation and confusion

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Fentanyl Patch

- Dangers:
  - Avoid in opioid naive patients
    - Fatal respiratory depression
  - Too rapid titration
    - Start with lowest strength
- 24-72 hours to reach maximum effect
- Heat increases drug release
- Change every 3 days



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Fentanyl analogues

- Alfentanil, Remifentanyl & Sufentanil
  - More rapid onset: 1-2 minutes
  - Shorter duration of action
  - Remifentanil useful in multi-organ failure
  - Conscious sedation

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Use of Opioid patches

- Apply to dry, cool, hairless skin
  - Upper chest/outer arm
- Write date/time of application
- Remove old patch before applying new one
- Avoid excess body heat e.g. saunas
- Remind health professionals you are wearing a patch e.g. On admission to hospital
- After use....fold patch on itself and dispose of carefully

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Tramadol



- MOA
  - Synthetic analgesic
    - weak mu agonist effect
    - some inhibitory action on Noradrenaline and serotonin reuptake in the CNS.
    - Potent metabolite
  - Its analgesic effects thought to be independent of mu receptor; only partially antagonised by Naloxone

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Tramadol: Adverse effects

- Common: Nausea, vomiting, sweating, sedation & dizziness.
- More serious: Confusion, hallucinations and seizures (care in epilepsy).
- Serotonin syndrome.
  - Serotonin poisoning.
  - Mental state changes: confused, agitated.
  - Tremor, hyperreflexia.
  - Shivering, sweating, fever.
- Drug interactions: TCA's, SSRI's and warfarin.

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Methadone

- VERY strong bond, means that it knocks off heroin or morphine
- Useful in preventing a good portion of the withdrawal symptoms
- Very long half life.

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Methadone





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Methadone

- NMDA receptor antagonist
  - Mu receptor agonist
- Long half life; dose titration difficult
  - 15-100 hours
  - 12 days after a dose change
- No toxic/active metabolites
- High bio-availability
- ADR's: sedation and confusion
- Drug interactions

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Methadone

- Difficult to dose adjust
  - Single dose: 1-2 as potent as morphine
  - Long term: 10 times as potent
- Used in Palliative care patients
  - Given twice a day
- Stigma
- SSRI's….toxicity

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Opioid Potencies

Compared to oral morphine

Codeine	0.1	3-6 hours.
Tramadol	0.2	4-6 hours.
Oxycodone	1.5-2.0	3-4 hours.
Hydromorphone	7.5	4-5 hours.
Methadone	5-10	8-11 hours
Fentanyl patch	100-150	72 hours.

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Analgesics+ : Part 4
Neuropathic and Chronic Pain / Anaesthesia

Introductory Pharmacology PHA00315



Ken Wojcikowski, PhD (Pharmacology and Toxicology; School of Medicine, University of Queensland)

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Specific objectives for Part 4

- Define chronic pain and neuropathic pain
- Describe the medications used to treat chronic and neuropathic pain
  - *Amitriptyline*
  - *Gabapentin*
- Name the 4 stages of anaesthesia
- Briefly state the indications for Bupivacaine.

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Understanding chronic pain and neuropathic pain

Dr Ken Wojcikowski

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Neuropathic Pain

- Drug, disease or injury induced damage or dysfunction of nervous system
- Burning, cold, tingling, numbness
- Common in:
  - Phantom limb pain
  - Spinal cord injury
  - Peripheral neuropathy
  - Post herpetic neuralgia
  - Cancer related pain
  - Diabetic neuropathy

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Management of chronic and neuropathic pain

Treat aggressively and early

1. Start with TCA or SNRI
2. Trial one drug at a time
3. Combination approach
4. Assess effectiveness
5. Balance efficacy, cost & adverse effects

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Adjuvant Analgesics

- Corticosteroids
- Antidepressants
- NMDA receptor channel blockers
- Benzodiazepines: *alter perception of pain*
  - Short term use
  - Muscle relaxant
- Anticonvulsants
  - Sodium Valproate
  - Carbamazepine
  - Gabapentin
  - Lamotrigine ineffective

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Tricyclic Antidepressants

- E.g. *amitriptyline*, nortriptyline
- Neuropathic pain…smaller doses
- Drowsiness, dry mouth, constipation, impaired cardiac conduction
- Postural hypotension….elderly. Less with nortriptyline
- Doses above 75mg: adverse effects
- Response depends on duration of pain
- Opioid sparring in cancer pain

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Carbamazepine

- Stabilise membranes/inc effectiveness of GABA
- Start slow and titrate up. 50-100mg od-bd
- Nausea, dizziness, rash and sedation
- Take with food
- Monitor liver, renal and blood counts
- Enzyme inducer and increases metabolism of tramadol and methadone
- First choice in trigeminal neuralgia

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Gabapentin & Pregabalin

- Interfere with release NA, substance P & glutamate
- Start low and titrate up
- Renally excreted
- Dizziness, somnolence and ataxia
- Neuropathic pain 900-3600mg/day
- No drug interactions
- Short half life
- PBS

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Gabapentin/Pregabalin

- Gabapentin: Useful in post herpetic neuralgia & diabetic neuropathy
- Pregabalin:
  - Limited data
  - Anticonvulsant, analgesic
  - Inc dose slowly. Start at 75mg
  - Renally excreted unchanged
  - Users..as above

- *Cochrane review 2011: High level pain relief in a third of patients. Moderate benefit in 50% of patients. NNT 6.*
  - Frequent & tolerable side effects.

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Neuropathic pain: Practise Points

- Can combine adjuvant analgesics
- Non drug strategies e.g. physiotherapy, sleep hygiene
- Topical capsaicin 0.75%
- Avoid:
  - Two different antidepressants
  - An antidepressant with tramadol…serotonin toxicity

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Marijuana and neuropathic pain

Marijuana is beneficial for relief of neuropathic pain

- Purified extracts of certain cannabinoids or of whole plant extracts consumed orally will be found to be most beneficial
- Teas will be found to be least effective
  - Cannabinoids are oil based substances and very little will be available in tea
- Smoking will be found to have health risks such as lung problems and sudden 'peaks'

Standardised doses needed for safety

Contraindicated in persons under 21 years of age

Check local laws regarding legality

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To date, the TGA has approved applications for the use of marijuana for the following indications:

- chemotherapy-induced nausea and vomiting
- refractory paediatric epilepsy
- palliative care indications
- cancer pain
- neuropathic pain
- spasticity from neurological conditions
- anorexia and wasting associated with chronic illness (such as cancer).

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A word about anaesthesia

Dr Ken Wojcikowski

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Anaesthetics

Pharmacology

- Prevents repolarisation/block electrical impulses:
  - blocks Na channels
- Reversibly interrupts conduction
- High lipid solubility….more potent
- Lipophilic head/hydrophilic tail



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History of Anaesthesia

- Church "Anaesthesia is a violation of god."
  - Belief that inflicted pain strengthened faith and taught the new mother of the need of sacrifice for her children.
- Church " In sorrow, thou shalt bring forth children."
  - Patients thought anaesthesia was a divine blessing.
  - Concerns re ethics of operating on a comatose patient.
- Queen Victoria: 1853 and 1857.

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Amputation without anaesthesia



Amputation, Thomas Rowlandson, 1793.

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Opioids in the 19th Century

- 1800's standard treatment for acute pain.
- Morphine industrially produced in Germany 1820's.
- 1855 Alexander Wood devised a syringe with a hollow needle…overuse.
- Opium and alcohol based compounds unregulated and available over the counter.
- 1870's….concern about morphine habit.
- 1890's di-acetylated morphine to produce Heroin.
- Bayer..development of aspirin.

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General anaesthetics

- Act in the CNS
- Activate GABA and/or glycine receptors
- Suppress NMDA receptors
  - Inhalation anaesthetics:
    - Nitrous oxide
    - Methoxyflurane, sevoflurane

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Stages of anaesthesia

Stage 1: Analgesia without amnesia

Stage 2: Excitement (nausea, hyperactivity, irregular respiration)

Stage 3: Surgical anaesthesia (sleep, normal respiration

Stabe 4: Medullary depression (coma, death)

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Local anaesthetics

- Long acting: More protein bound
  - Bupivicaine..3-10 hrs. PCA/epidural post op
  - Potential for cardiotoxicity
  - Major nerve blocks, epidurals, spinals etc
  - Ropivicaine….less cardiotoxic
    - Minor and major nerve blocks, epidurals and post operative pain
- Short acting:
  - Lignocaine…duration increased with adrenaline

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Local anaesthetics

- Add adrenaline
  - Reduce/slow peripheral absorption
  - Lowers peak blood concentration: less toxicity
  - Intensifies local anaesthetic effect
- pH adjustment
  - sodium bicarbonate 8.4%: reduce stinging
  - Increases rate of nerve block

EMLA:
Eutectic Mixture of Local Anaesthetics

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Bupivacaine

- Injected around a nerve that supplies the area
- Mixed with a small amount of adrenaline
  - Increases the duration of its action
- Blocks sodium influx into nerve cells, which prevents depolarization.
- Lasts for 2 to 8 hours



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Useful resources

- AMH (only one subscription in library)
- AusDI (unlimited students can enter)
- Therapeutic Guidelines Analgesia
- www.australianprescriber.com.au
- Ward Pharmacist

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References (pictures for entire topic)

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